

Quab. E₄

$$u = ax^2 + bxy + c,$$

$$v = by^2 + cx + mz,$$

$$w = mz^3.$$

$$e_{ij} = \frac{1}{2} \left[\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j} \right]$$

$$\frac{\partial^2 e_{xx}}{\partial y \partial z} = \frac{\partial}{\partial x} \left(-\frac{\partial e_{yz}}{\partial x} + \frac{\partial e_{zx}}{\partial y} + \frac{\partial e_{xy}}{\partial z} \right),$$

$$\frac{\partial^2 e_{yy}}{\partial z \partial x} = \frac{\partial}{\partial y} \left(-\frac{\partial e_{zx}}{\partial y} + \frac{\partial e_{xy}}{\partial z} + \frac{\partial e_{yz}}{\partial x} \right),$$

$$\frac{\partial^2 e_{zz}}{\partial x \partial y} = \frac{\partial}{\partial z} \left(-\frac{\partial e_{xy}}{\partial z} + \frac{\partial e_{yz}}{\partial x} + \frac{\partial e_{zx}}{\partial y} \right),$$

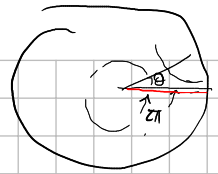
$$2 \frac{\partial^2 e_{xy}}{\partial x \partial y} = \frac{\partial^2 e_{xx}}{\partial y^2} + \frac{\partial^2 e_{yy}}{\partial x^2},$$

$$2 \frac{\partial^2 e_{yz}}{\partial y \partial z} = \frac{\partial^2 e_{yy}}{\partial z^2} + \frac{\partial^2 e_{zz}}{\partial y^2},$$

$$2 \frac{\partial^2 e_{zx}}{\partial z \partial x} = \frac{\partial^2 e_{zz}}{\partial x^2} + \frac{\partial^2 e_{xx}}{\partial z^2}.$$

Quia 6, Ex 5

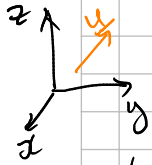
$$\theta = [0, 2\pi]$$



$$u = ar \log \theta,$$

$$v = ar^2 + c \sin \theta,$$

$$w = 0.$$



$$\underline{u}(r, \theta) = \begin{bmatrix} u_x \\ u_y \\ u_z \end{bmatrix} = \begin{bmatrix} u(r, \theta) \\ v(r, \theta) \\ w(r, \theta) \end{bmatrix}$$

$$\epsilon_{rr} = \frac{\partial u_r}{\partial r}$$

$$\epsilon_{\theta\theta} = \frac{u_r}{r} + \frac{1}{r} \frac{\partial u_\theta}{\partial \theta}$$

$$\epsilon_{r\theta} = \epsilon_{\theta r} = \frac{1}{2} \left(\frac{1}{r} \frac{\partial u_r}{\partial \theta} + \frac{\partial u_\theta}{\partial r} - \frac{u_\theta}{r} \right)$$



$$\underline{u}(r, \theta) = \begin{bmatrix} u_r \\ u_\theta \\ u_z \end{bmatrix}$$

$$u_r = u \cdot \cos \theta + r \cdot \sin \theta$$

$$u_\theta = -u \cdot \sin \theta + r \cdot \cos \theta$$

$$e_{rr} = \frac{\partial u_r}{\partial r},$$

$$e_{\theta\theta} = \frac{u_r}{r} + \frac{1}{r} \frac{\partial u_\theta}{\partial \theta},$$

$$e_{r\theta} = \frac{1}{2} \left(\frac{\partial u_r}{\partial \theta} + \frac{\partial u_\theta}{\partial r} - \frac{u_\theta}{r} \right),$$

$$e_{rz} = \frac{1}{2} \left(\frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r} \right),$$

$$e_{z\theta} = \frac{1}{2} \left(\frac{\partial u_z}{\partial \theta} + \frac{\partial u_\theta}{\partial z} \right),$$

$$e_{zz} = \frac{\partial u_z}{\partial z}.$$

$$S_{zz} \equiv \frac{2}{r} \frac{\partial^2 \epsilon_{\theta r}}{\partial \theta \partial r} - \frac{\partial^2 \epsilon_{\theta\theta}}{\partial r^2} - \frac{1}{r^2} \frac{\partial^2 \epsilon_{rr}}{\partial \theta^2} + \frac{1}{r} \frac{\partial \epsilon_{rr}}{\partial r} + \frac{2}{r^2} \frac{\partial \epsilon_{r\theta}}{\partial \theta} - \frac{2}{r} \frac{\partial \epsilon_{\theta\theta}}{\partial r} = 0.$$